

Syllabus for Mid Semester Examination for Advanced Machine Learning

1. Machine Learning Project Lifecycle (Part 1)
2. Machine Learning Project Lifecycle (Part 2)
3. Simple Linear Regression using OLS Method
4. Multiple Linear Regression using OLS Method
5. Batch Gradient Descent
6. Stochastic Gradient Descent and Mini Batch Gradient Descent
7. Evaluation Metrics (Regression and Classification)
8. Polynomial Regression and Assumptions of Linear Regression
9. Bias, Variance and their Tradeoff
10. Feature Selection
11. Principal Component Analysis
12. Regularization
13. Time Series Analysis
14. Maximum Likelihood Estimation and Logistic Regression
15. Gradient Descent on Logistic Regression and Multiclass Classification